Passing Coarse Marginal Checks Can Be Cheap: Persona Mixtures and Imprecise Treatment-Response Estimates in an LLM Persona Panel

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July 2026

Author: Yohei Nakajima (Untapped Capital). Experiments executed by an autonomous pipeline (Replit Agent + ActiveGraph event-sourced engine). Attribution and reviewer-role disclosure: §8.

Artifacts (public): github.com/yoheinakajima/synthetic-players — anonymous clone and one-command zero-credential verifier; 4,916 Phase 3–5 runs verified (4,896 LLM runs replayed byte-exact plus 20 deterministic baselines independently recomputed); prompt registries and freeze records are externally anchored.

Abstract

Large language models are increasingly used as synthetic research participants, but they are often validated by whether their marginal responses resemble published human data. We report a five-phase research program whose confirmatory claims from Phases 3–5 were registered before their adjudicating data and mechanically evaluated from an event-sourced record. A fixed panel of sixteen lightweight persona-conditioned GPT-4.1 configurations passed preregistered broad-reference checks for condition-level cooperation in three of four repeated-game cells. A fixed-panel Dirichlet–Jeffreys latent-propensity sensitivity yields posterior median between-prompt shares of 63%–71%, with 95% intervals spanning 49%–81%; finite-opportunity plug-in estimates that condition more strongly on the observed boundary concentration are 85%–96%. The observed aggregate continuation-probability contrasts are +0.083 and +0.078 across two wording families, with conservative exact simultaneous 95% intervals [−0.171, +0.330] and [−0.181, +0.330] (rounded to three decimals). Their width jointly reflects six independent episodes per prompt-cell and an exact construction that retains uncertainty at empirical corners. The treatment changed both the continuation process and the language communicating it, leaving incentive and framing channels undecomposed. Separate representation experiments showed that, in the bare configuration, replacing and repositioning one continuation sentence shifted cooperation from 0/40 to 37/40 on held-out decisions, and that a displayed action label or label-linked learned prior could override payoff dominance in one registered conflict cell. External review exposed family-error, dependence, and boundary-uncertainty defects; zero-call reanalysis changed the scientific interpretation without rewriting the historical record. The public capsule now verifies all 4,916 Phase 3–5 runs with no live model calls. Human references are protocol-nonmatched, the results concern one fixed model–persona panel, and we do not claim human substitutability.

1. Introduction

Human behavioral experiments are slow and expensive; LLM calls are fast and nearly free. A growing literature reports that suitably conditioned LLMs produce data resembling human data—“algorithmic fidelity” [Argyle et al. 2023], “homo silicus” [Horton 2023], behavior “statistically indistinguishable from a random human” [Mei et al. 2024]—and a formal framework for evaluating *statistical realism* now exists [Xie et al. 2026]. Much of this evidence validates marginals: means, distributions, and aggregate replication.

Recent work shows that descriptive realism and causal fidelity can diverge at scale [Li & Ji 2026], that treating LLM outcomes as surrogates for human outcomes requires assumptions that marginal equivalence does not supply [Persson et al. 2026], and that intervention prompts can shift a model's implied latent user even when explicit persona text is fixed [Lin et al. 2026]. Li and Ji additionally trace effect errors to intervention logic, outcome structure, and excessive attitude–behavior coupling, so the present paper does not claim mechanism-level explanation in general. Its narrower contribution is a specific fixed-panel composition mechanism in incentive-bearing strategic interaction, coupled to minimal representation interventions and an auditable correction record. We decompose the explicit persona panel into between-prompt dispersion, within-prompt variation, and prompt-indexed treatment response. The panel passes coarse marginal checks while producing small but imprecisely estimated continuation-probability contrasts, and most recorded variation lies between prompt configurations rather than within them. Sealed templates and matched procedures control explicit assignment, environment randomization, and execution. They do not establish latent-person invariance; Lin-style user drift and the observed composition pattern can coexist.

Here, "cheap" denotes evidentiary economy rather than only low API cost: the broad-reference marginal checks could be passed without estimating the treatment-response object they might be taken to validate.

Contributions. First, we provide a registered strategic-interaction example in which a fixed persona panel passes coarse condition-level and dispersion checks while aggregate continuation-probability point differences remain imprecisely estimated; the design does not establish equivalence or a narrow response bound (§4.1). Second, we quantify the associated composition problem with three explicitly different uncertainty views: a fixed-panel latent-propensity posterior, finite-opportunity plug-in estimates, and an exploratory persona-generator bootstrap; the first places median between-prompt shares at 63%–71%, while representation experiments show how wording and displayed labels or learned game priors govern the induced policies (§4.1–4.3). Third, we demonstrate an auditable reliability protocol—and its limits—through prospective registration, external chronology, mechanical adjudication, complete zero-call replay of the confirmatory record, and public correction of family-error and construct-validity defects (§4.4–4.5).

2. Related work

Occupied territory, and where we sit. Li and Ji [2026] establish across three model families, eleven interventions, and 59,508 participants that descriptive fit and intervention-effect accuracy can diverge, that prompt refinements improving realism do not reliably improve effect accuracy, and that errors vary with intervention logic, outcome structure, and attitude–behavior coupling. Persson, Schultzberg, and Ankargren [2026] formalize when LLM outcomes can serve as causal surrogates and why novel interventions still require human evidence. Lin et al. [2026] show that interventions can change the implicit simulated population even when explicit personas are fixed. Statistical-realism, persona-collapse, and state-versus-trait work further show that persona-conditioned populations can compress or misallocate heterogeneity [Xie et al. 2026; Harry et al. 2026; Xiao et al. 2026]. Our differentiation is therefore not the broad divergence or the existence of mechanisms. It is a registered decomposition of one common lightweight construction in strategic interaction: the same fixed explicit prompt panel is evaluated for marginal fit, finite-opportunity-corrected between/within composition, representation sensitivity, and response to a represented continuation-probability treatment, with exact prompt provenance and public inferential correction. The observed composition pattern is complementary to, not exclusive of, latent-user drift.

LLM strategic behavior. Akata et al. [2025] characterize repeated-game play modulated by prompts; Pal et al. [2026] elicit strategies from five models while varying continuation probability, payoffs, horizon knowledge, and framing; counterfactual-reasoning evaluations alter labels and payoff structures [Georgousis et al. 2026]; and "strategic robustness" has been defined as payoff-preserving invariance across narratives [Mousavi Davoudi et al. 2026]. These works establish that neither repeated games nor prompt/payoff perturbations are new. Our distinct combination is the fixed persona panel, the explicit between/within/response decomposition, exact prompt provenance, prospective registration of confirmatory claims, and mechanical adjudication followed by public inferential correction.

Strong positive evidence, and a different estimand. Ashokkumar, Hewitt, Ghezae, and Willer [2026] use study descriptions to forecast 469 effects from 70 preregistered, nationally representative survey experiments and find strong correlations with realized effects, alongside systematic effect-size overestimation and weaker performance in a megastudy archive. That is important contrary evidence against any blanket pessimism about LLMs in experimental science. It is also a forecasting task over studies rather than subject-level simulation of a response surface. Strong effect forecasting is compatible with the fixed-panel composition failure studied here and reinforces the prescription to validate a simulator on the exact response object for which it will be used.

Synthetic participants and personas. Bisbee et al. [2024] find plausible survey averages alongside compressed variance, distorted coefficients, and temporal drift; Boelaert et al. [2025] report excess homogeneity; Anthis et al. [2025] catalog diversity and generalization challenges; Hullman et al. [2026] propose statistical calibration for confirmatory use; and Park et al. [2024] show that rich interview conditioning can substantially outperform lightweight demographic/persona descriptions. Format sensitivity [Sclar et al. 2024], role-play framing [Shanahan et al. 2023], persona collapse [Xiao et al. 2026], state blindness [Harry et al. 2026], and reviews of persona-experiment transparency [Batzner et al. 2025] all caution against treating a persona string as a stable human analogue. RLHF-related diversity reduction is a possible mechanism for concentrated policies, not a mechanism identified by this design. Full map: `docs/analysis/literature-map.md`; differentiation table: `docs/analysis/novelty-relationships.md`.

3. Instrument and inferential units

The primary deployment is gpt-4.1 with 16-token outputs and a fixed minimal behavioral-subject prompt containing no game-theory vocabulary or reasoning scaffold. Temperature was 0.7 except in the registered Phase 5 sweep at 1.0 and 1.3. On the primary OpenAI-compatible path, `temperature` and `max_tokens=16` were explicitly supplied; the assembled prompt set `top_p=1.0`, which the adapter intentionally omitted from the wire at 1.0, while `presence_penalty`, `frequency_penalty`, and `logit_bias` were not supplied and therefore inherited provider defaults. No tools or native structured output were used. Phase 5 prepends one sealed persona sentence to byte-identical task text. The cross-vendor Gemini tier is descriptive; the original Claude Haiku candidate failed a registered entry gate and was replaced under an archived amendment. Environment randomness is seeded; provider-side generation is not claimed to be seeded. Every request, rendered prompt, completion, decoding configuration, round, and provenance record is archived.

3.1 Sequential architecture and registration status

Stage	Primary question and role	Unit used in the paper	Registration status
Phase 1	Initial prototype and naive behavioral claims; establishes the historical baseline, not current confirmatory evidence	provider calls / recorded decisions	post hoc instrument development

Stage	Primary question and role	Unit used in the paper	Registration status
Phase 2	Mechanical re-adjudication and enforcement repair after the initial harness exposed analyst discretion	archived runs and claim predicates	corrective, not prospective confirmation
Phase 3	Bare GPT-4.1 configuration in repeated PD, framing, and RPS	episode, with historical seat-level summaries disclosed	claims registered before Phase 3 data
Phase 4	Representation robustness, X1/X2 wording extensions, counterfactual payoffs/labels, continuation-probability assays, adversaries, and sentinels	complete episode for current sensitivities	X1 was a sequentially registered, result-informed extension: Phase 3 motivated the test, while its prompts, sample size, and predicate were sealed before any X1 data; the remaining blocks were registered before their own data
Phase 5	Sixteen sealed persona-prefix configurations crossed with the Phase 4 instruments; descriptive Gemini tier	complete persona prompt for the fixed panel; episode beneath it	confirmatory predicates registered before Phase 5 data; post-adjudication sensitivities are explicitly unregistered

The present paper's main empirical decomposition is Phase 5, interpreted using representation results from Phases 3–4. Phases 1–2 document instrument evolution and are not counted as prospective confirmation.

The full event store contains 5,505 completed runs, 54,276 round events, 108,552 seat-round decisions, and 36,251 archived provider-request events. The public confirmatory replay contract now verifies 4,916 Phase 3–5 runs: 320 registered Phase 3/X1 LLM runs, 20 deterministic Phase 3 baselines, 2,864 Phase 4 runs, and 1,712 Phase 5 runs; three additional completed legacy entry/diagnostic runs are also replayed but are not counted as confirmatory. A separate transactional ledger records 30,530 Phase 4–5 calls, 13,141,675 input tokens, and 45,247 output tokens; it excludes earlier phases and therefore must not be conflated with the full event-store request count. Counts and definitions are reconciled in `docs/analysis/submission/count-reconciliation.md`.

Confirmatory claims were registered before the data that adjudicated them and were mechanically evaluated in a fixed vocabulary. The historical two-sided interiority rule used Clopper–Pearson bounds on seat-level round-one trials. Because two seats share an episode, the submission analysis additionally treats the complete episode as the independence unit. For an episode outcome $Y \in \{0, 0.5, 1\}$, the conservative exact sensitivity writes

$$Y = \tfrac{1}{2}\{\mathbf{1}(Y \geq 0.5) + \mathbf{1}(Y = 1)\},$$

constructs simultaneous Clopper–Pearson intervals for the two episode-level binary components, and projects them onto $E[Y]$. For wording family w, prompt i, continuation condition d, and six complete episodes e, define $A_{ide} = \mathbf{1}(Y_{ide} \geq .5)$ and $B_{ide} = \mathbf{1}(Y_{ide} = 1)$, so $\widehat{p}_i(d) = \{\bar{A}_i(d) + \bar{B}_i(d)\}/2$. For each aggregate contrast, the sixteen equally weighted prompt cells contribute 96 episodes per condition; pooled component counts therefore estimate $\bar{p}(d) = 16^{-1} \sum_i p_i(d)$. The four component-condition intervals split the total error rate by Bonferroni. If $[L_d, U_d]$ is the projected condition-mean interval, the contrast interval is

$$[L_{.90} - U_{.10},\ U_{.90} - L_{.10}].$$

Under the pooled independent-binomial component model this union-bound construction has at least 95% simultaneous coverage; with independent heterogeneous prompt probabilities the component totals are Poisson-binomial and no more dispersed than the equal-probability binomial at the same mean, making the binomial projection conservative for that working model; its role is a conservative small-sample projection rather than a claim that the sixteen prompt propensities are homogeneous. It does not assume seat independence and does not collapse to zero uncertainty when all observed episodes agree. A Dirichlet–Jeffreys sensitivity uses the symmetric Dirichlet(0.5, 0.5, 0.5) prior on the probabilities of episode outcomes {0, 0.5, 1}; posterior draws project $E[Y] = 0.5q_{0.5} + q_1$. The percentile cluster bootstrap is also retained as a post-adjudication sensitivity. The exact Clopper–Pearson projection is the conservative reference because it provides finite-sample coverage for the discrete episode mean; at n=6 the percentile bootstrap has no comparable coverage guarantee and can understate uncertainty. Its degeneracy at exact corners is a symptom of that limitation, not by itself a false-positive mechanism for this strict interiority gate.

The hierarchy is deployment → explicit persona prompt → condition → episode → seat → round → provider request. Phase 5's confirmatory unit is the complete persona sentence; name, age, occupation, and traits are bundled semantic treatments. Registered claims attach to the conditional finite-panel estimand for these sixteen prompts. Claims about a wider persona generator are exploratory at $n = 16$. Pairing the same explicit prompt across conditions identifies a prompt-indexed contrast, not necessarily a stable latent person's treatment effect.

Protocol glossary. `S2-absent` and `S2-present` are the registered repeated-game wording families. **switch-bearing** means the span whose adjacent substitution produced the largest preregistered ladder gap and subsequently passed held-out confirmation; S2-present contains that replacement-and-reposition operation, while S2-absent contains the original sentence. `P3-A3` is the Phase 3 registered broad-reference cooperation claim, with band [0.36, 0.63]. `P5-1a`, historically called the **corner-mixture predicate**, is the registered support condition that fires when the interior fraction in the exact-bare-twin restricted set is below 0.10 under the frozen seat-level rule; it is not a general theorem about mixture structure. `P5-1b` is the registered between-persona dispersion comparison. `P5-2` pools registered conflict cells and classifies whether choices follow task text or persona-conditioned direction. `P5-3(a)`—clause (a)—asks whether any persona × wording pair has both continuation-probability cells interior and a positive slope lower bound; `P5-3(b)`—clause (b)—asks whether each persona lane rejects the bare configuration's dominated swap-cell option at a registered minimum rate. Historical verdict labels remain visible even where post-adjudication analyses change their scientific interpretation.

Phase 5 condition matrix. The 96 Tier-A persona–condition units are the full cross of sixteen prompts with six conditions:

Code	Condition	Role in the paper
`rep-d10-s2a`	repeated PD, δ=.10, S2 absent	repeated-game level, variance, and response

Code	Condition	Role in the paper
`rep-d10-s2p`	repeated PD, δ=.10, S2 present	repeated-game level, variance, and response
`rep-d90-s2a`	repeated PD, δ=.90, S2 absent	repeated-game level, variance, and response
`rep-d90-s2p`	repeated PD, δ=.90, S2 present	repeated-game level, variance, and response
`os-swap`	one-shot canonical-payoff label swap	semantic-label/payoff conflict
`os-community`	one-shot Community framing	near-interior framing anchor

The registered P5-1a concept restricted its primary denominator to persona cells whose **exact recorded bare twin** failed the same interiority gate. An outcome-blind exact-twin completion fixed that set as `rep-d90-s2a` and `os-swap`: sixteen personas in each condition, hence 32 units. The exact Community twin passed the bare gate; the other three repeated-game cells lacked exact bare twins and entered only the unrestricted 96-cell secondary.

The machinery's boundary is explicit:

> **The pipeline can enforce a registered predicate exactly; it cannot guarantee that the predicate represents a valid estimand, test family, or construct.**

4. Results

4.1 Coarse marginal checks pass while represented-treatment estimates remain imprecise

The preregistered leaning rule—at least two of agreeable, patient, and risk-averse—divides the fixed panel into eight cooperative-leaning and eight defect-leaning complete prompts. The descriptive gaps range from 0.510 to 0.719 across every non-swap condition:

condition	cooperative-leaning mean	defect-leaning mean	difference	prompts per stratum
`rep-d10-s2a`	0.615	0.083	+0.531	8
`rep-d10-s2p`	0.760	0.094	+0.667	8
`rep-d90-s2a`	0.688	0.177	+0.510	8
`rep-d90-s2p`	0.865	0.146	+0.719	8
`os-community`	0.688	0.019	+0.669	8

These are fixed-panel prompt-bundle contrasts, not causal estimates for any trait. The P3-A3 broad-reference cooperation band is [0.36, 0.63]. The four repeated-game pool means are 0.349 (`rep-d10-s2a`), 0.427 (`rep-d10-s2p`), 0.432 (`rep-d90-s2a`), and 0.505 (`rep-d90-s2p`); only the S2-absent δ=.10 cell falls outside the band, by 0.011 below its lower boundary.

P5-1b used protocol-nonmatched human SD references mechanically implied from Dal Bó and Fréchette's [2011] R=40 strategy-frequency estimates: 0.4122 for their δ=.50 panel and 0.3116 for δ=.75. The frozen ratio ρ=.75 gives thresholds 0.3092 and 0.2337. The ratio was a preregistered heuristic tolerance—not a theoretically derived equivalence or psychometric margin—so P5-1b is interpreted as a permissive historical dispersion checkpoint rather than evidence of human-variance equivalence. Finite-opportunity-corrected plug-in SDs are 0.4182, 0.4784, 0.4408, and 0.4323, and all exceed the historical thresholds.

The uncertainty-propagating fixed-panel view is more conservative. Independent Dirichlet(0.5,0.5,0.5) posteriors for each prompt/cell outcome distribution yield between-prompt-share medians of 63.1%, 70.5%, 66.1%, and 66.5%, with 95% intervals [49.4%, 74.5%], [57.3%, 81.3%], [52.8%, 77.0%], and [52.8%, 77.7%]. The corresponding latent-SD medians are 0.3543, 0.3959, 0.3698, and 0.3620. Three of four lower bounds exceed the historical SD threshold; `rep-d10-s2a` is effectively on it (0.3090 versus 0.3092).

For comparison, finite-opportunity plug-in shares are 85.5%, 96.1%, 88.8%, and 90.2%, with conditional episode-bootstrap 95% intervals [82.0%, 93.8%], [94.6%, 98.9%], [86.7%, 94.6%], and [87.9%, 95.5%], respectively. A conditional episode bootstrap that resamples the empirical distribution of each fixed prompt produces corrected-SD intervals [0.4122, 0.4391], [0.4696, 0.4916], [0.4279, 0.4654], and [0.4269, 0.4496], but empirically unanimous cells remain point masses in that bootstrap. The visually suspicious first lower bound is genuine: the stored full-precision value is 0.412198; an independent implementation that imports neither the original variance routine nor human constants produced 2.5th-percentile bounds from 0.412128 to 0.412128 across three 250,000-replicate seeds. Its equality to the displayed human reference at four decimals is a rounding coincidence, not a computational link. A two-stage prompt+episode bootstrap changes the estimand toward a hypothetical persona generator and yields still wider SD intervals: [0.2724, 0.4879], [0.3696, 0.5123], [0.3457, 0.4890], and [0.3345, 0.4847]. Thus the robust statement is that between-prompt composition is substantial and likely dominant in this fixed panel; 85%–96% are conditional plug-in point estimates, not fully uncertainty-adjusted population facts.

Across the represented continuation-probability treatment, the observed fixed-panel point differences are +0.083 for S2-absent wording and +0.078 for S2-present wording. Conservative exact simultaneous 95% intervals are [−0.171, +0.330] and [−0.181, +0.330] (rounded to three decimals). Their breadth jointly reflects the registered six-episode-per-cell design and an exact projection that retains non-zero uncertainty at empirical corners. The treatment changes both the continuation process and the text used to communicate it; round-one actions identify response under a specified representation, not a semantically neutral economic parameter. The point estimates are small on the unit scale, but the data do not establish equivalence, a zero response, or a narrow upper bound.

For the finite archived panel, +0.083 and +0.078 are exact descriptive arithmetic. The intervals instead address repeated-sampling uncertainty about latent propensities. A design-effect heuristic using six episodes per prompt and the plug-in between-share range 0.855–0.961 gives roughly 16.5–18.2 episode equivalents per condition, close to the sixteen prompt units. This is not the degrees of freedom of the exact procedure, but it identifies prompt count as the operative precision constraint.

Dal Bó and Fréchette [2011] remain protocol-nonmatched context: their continuation probabilities, payoffs, monetary incentives, between-session assignment, and repeated-supergame experience differ from this study. We make no matched magnitude or human-equivalence claim.

The registered Gemini tier was descriptive and endpoint-nonstationary. Nine of 24 Gemini cells met the historical interiority rule, versus 14/96 in the primary panel, and several representation effects reversed direction. This is contrary descriptive evidence that the composition pattern is deployment-specific, not a formal replication comparison.

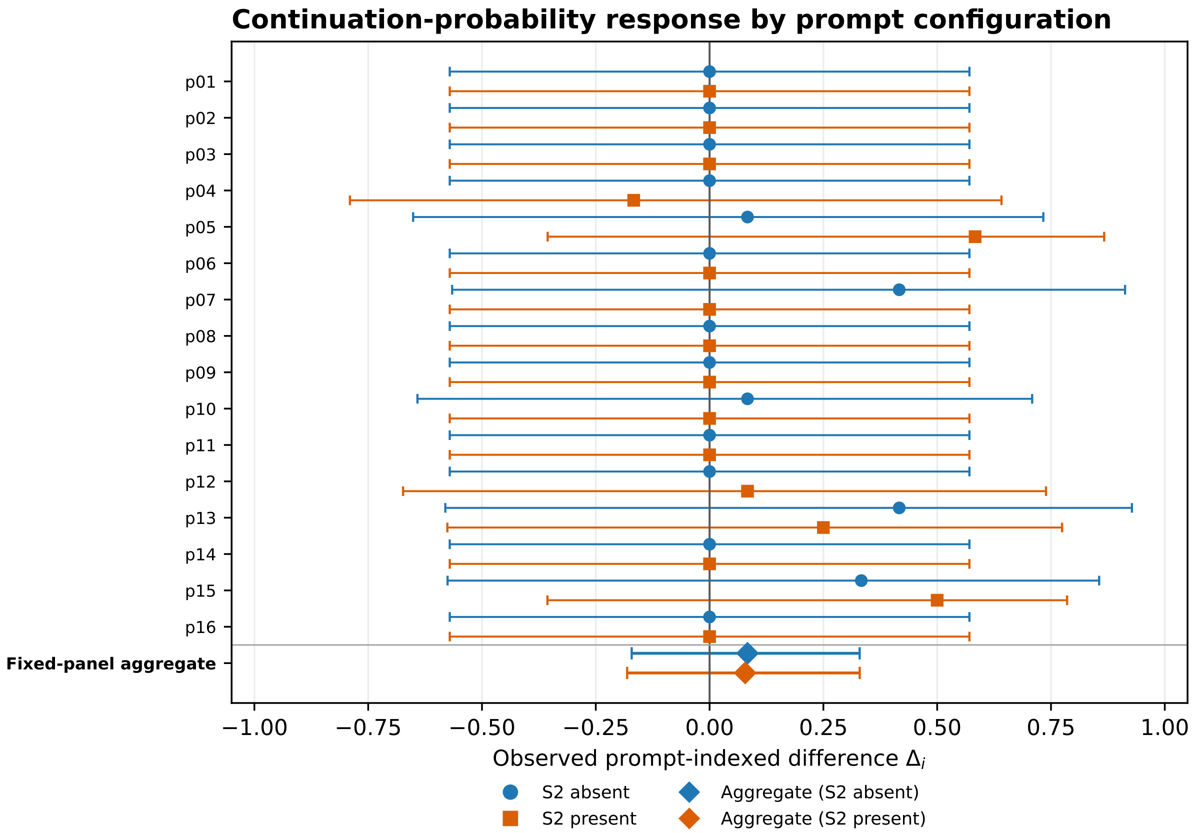

Figure 1. *Prompt-indexed differences in round-one cooperation, $\Delta_i = \hat{p}_i(\delta = .90) - \hat{p}_i(\delta = .10)$, for both wording families. Bars are conservative exact simultaneous 95% intervals with complete episodes as the unit; observed corners retain non-zero uncertainty. Blue and orange diamonds show the S2-absent and S2-present fixed-panel aggregates, respectively. Rows at $\Delta_i = 0$ can reflect boundary concentration in both recorded cells and are not precise evidence of no response.*

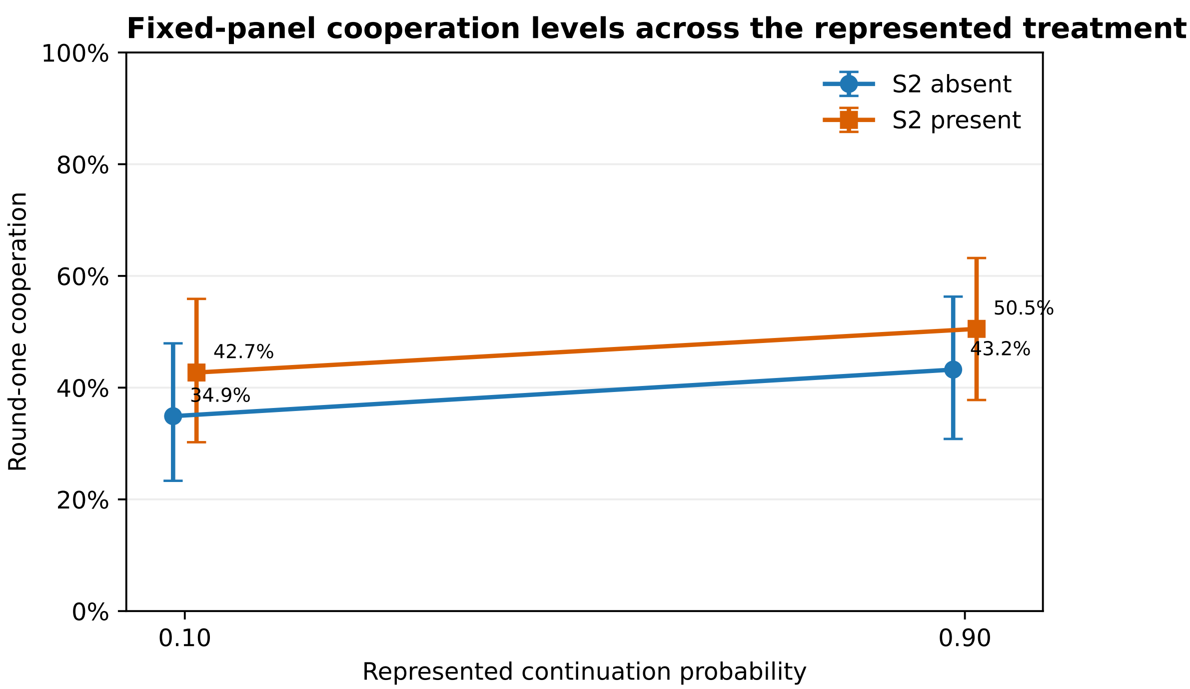

Figure 2. Fixed-panel cooperation by represented continuation-probability condition. Error bars are conservative exact condition intervals; lines connect conditions for orientation only.

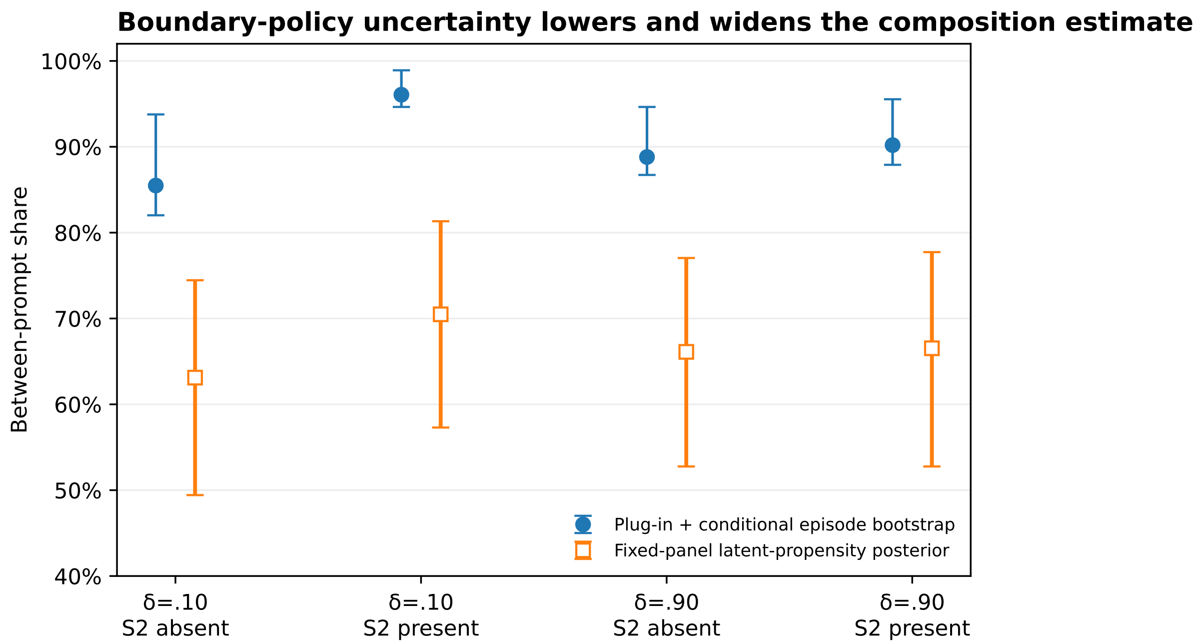

Figure 3. Between-prompt share of episode-level variation. The uncertainty-propagating Dirichlet–Jeffreys fixed-panel posterior is the primary interpretive sensitivity; plug-in/conditional-bootstrap estimates are shown as a complementary description of the archived concentration. The two-stage prompt bootstrap changes the estimand and is reported in text.

Across all six Phase 5 conditions, the historical seat-level rule classifies 14/96 persona–condition cells interior, the exact episode projection 11/96, and the Dirichlet–Jeffreys sensitivity 19/96. In the registered 32-unit set, the counts are 3/32, 2/32, and 5/32. At n=6 with a three-valued discrete outcome, modest differences in interval width deterministically move cells across the threshold; the divergence reflects both interval construction and low discrete-sample resolution. The continuous posterior and variance components are therefore more informative than the binary census.

4.2 What the marginal checks cannot identify

Let $p_i(d) = E[Y \mid i, d]$, where i indexes the complete explicit persona prompt and d the experimental condition.

Proposition A: broad bands only partially identify the aggregate contrast. If a synthetic condition mean is accepted within tolerance ϵ_d of a reference mean in each condition, then

$$|\Delta^S - \Delta^H| \leq \epsilon_0 + \epsilon_1.$$

Equivalently, accepted bands $[\ell_0, u_0]$ and $[\ell_1, u_1]$ imply only

$$\Delta^S \in [\ell_1 - u_0, u_1 - \ell_0].$$

Exact condition-specific mean matching would force the aggregate effect by the identity $\Delta = \mu_1 - \mu_0$; the empirical failure lives in the slack of coarse criteria.

Proposition B: aggregate moments do not identify microstructure or response coupling. This is an application of the law of total variance and classical Fréchet–Hoeffding/Sklar coupling results [Hoeffding 1940; Sklar 1959] to synthetic-participant validation, not a new probability theorem. Mean and total variance do not identify how variation is divided between prompt configurations and repeated draws, nor do they identify distributional shape or boundary concentration. Even exact condition-specific distributions do not identify the cross-condition coupling and therefore do not identify the distribution of prompt-indexed responses $\Delta_i = p_i(1) - p_i(0)$. Reusing an explicit persona string supplies one prompt-indexed coupling, but interpreting it as a stable synthetic individual's potential-outcome contrast requires latent-person invariance, which this study does not test.

The study therefore identifies a composition pattern in one fixed prompt panel. It does not establish that humans have a different microstructure, that RLHF caused the pattern, that latent-user drift is absent, or that the same pattern occurs across persona generators.

4.3 Control-channel interactions

For the bare configuration, round-one cooperation was 0.000 at every registered continuation probability. X2 decomposed the v1 and v2a prompt bundles into six sentence/block spans and constructed forward and reverse ladders by replacing one complete span at a time. The selected S2 operation replaced and repositioned "After every round there is a {deltaPct}% chance the session continues with another round" with "At the end of each round there is a {deltaPct}% chance that the session goes on for one more round." Screening used ten episodes per rung; the selected minimal pair was confirmed at temperature 0.7 on 20 fresh episodes per side (seeds 2953–2972), moving cooperation from 0/40 to 37/40. Because wording and position were one atomic operation, the design does not separate them.

In the label-swap cell, canonical payoffs were held fixed while "Cooperate" and "Defect" were attached to opposite strategic roles. The bare configuration chose the cooperation-worded option 0/40 times and instead took the payoff-dominated role whenever it carried the word "Defect." This shows that the displayed label or a label-linked learned prior can override payoff dominance in this registered cell. It does not identify intrinsic lexical valence: a learned game-theoretic association such as "Defect = equilibrium/dominant action in Prisoner's Dilemma" is an equally plausible mechanism. A structurally equivalent non-PD control retaining the same labels was not run.

Persona conditioning produces two observed contrasts, but they are not factorially separable. Differences among complete persona prompts generate the leaning gaps reported in §4.1. Adding any tested persona string reverses the bare swap-cell choice, yet no non-semantic prefix matched for length, punctuation, and position was run; semantic persona content cannot be isolated from generic sequence-format disruption. In the swap cell, label and payoff also point to the same option for persona-conditioned configurations, leaving the reversal mechanism ambiguous.

P5-3(b)'s 24 evaluable lanes comprise sixteen personas at T=0.7 plus p02, p06, p11, and p15 at each of T=1.0 and T=1.3. Every lane retains a simultaneous episode-exact lower bound above the frozen 0.20 threshold; the minimum is 0.462. The pooled P5-2 task-consistent share is 90/704 seat decisions across 352 episodes, equivalently 45/352=0.128 on episode means. The frozen historical adjudication used the seat-level rule. An episode-iid Clopper–Pearson projection gives [0.092, 0.172], but that interval does not propagate the prompt clustering visible elsewhere in the study. Two post-adjudication, zero-call sensitivities retain the historical point estimate while changing the uncertainty model: a stratified prompt-cluster bootstrap over the forty registered persona × conflict-cell clusters gives 95% interval [0.071, 0.189], and a fixed-panel Dirichlet–Jeffreys latent-propensity aggregation gives posterior median 0.172 with 95% interval [0.152, 0.195]. Both remain below the registered 0.20 persona-dominant boundary, although the Bayesian result approaches it. Every repeated conflict subcell is mixed; only the swap cell is individually persona-dominant. The pooled classification is therefore mechanism-confounded and carried by the swap cell, not evidence of a general persona-dominance mechanism.

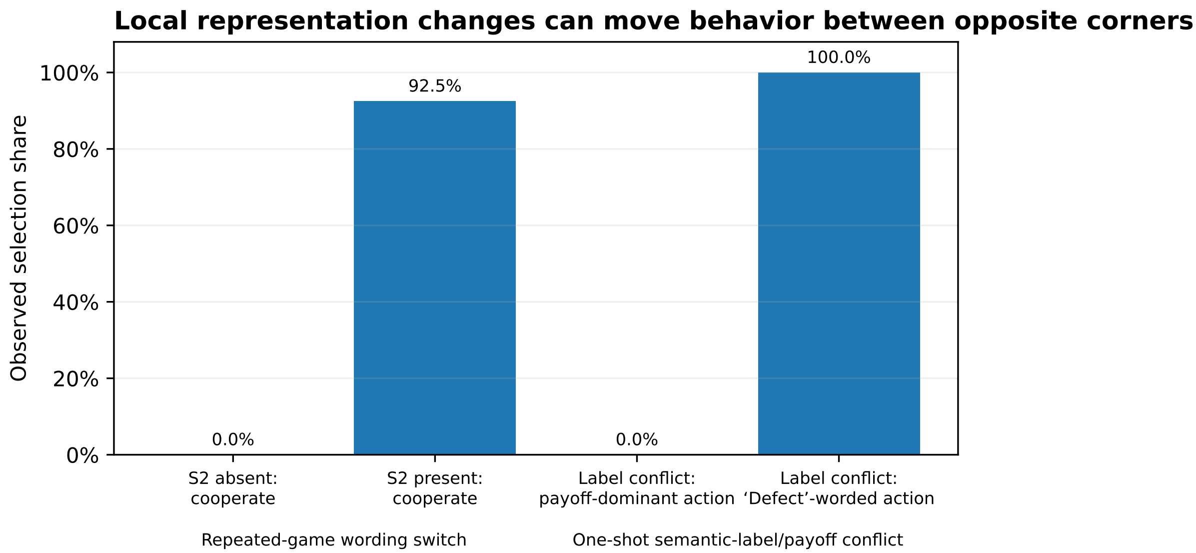

Figure 4. Two representation interventions in the bare configuration. The S2 wording-and-position operation moved cooperation from 0/40 to 37/40. In the one-shot label conflict, the payoff-dominant action was never chosen when the dominated role carried "Defect." These bars report selection shares, not a common or uniquely identified causal mechanism.

4.4 The favored persona-level result is not prospectively confirmed; the archived family is underpowered

Under the historical seat-level rule, persona p13 moved from 0.333 cooperation at δ=.10 to 0.750 at δ=.90 and passed a per-candidate lower-bound test. The rule searched multiple persona × wording candidates and fired on any pass without declared family-level error control. External review identified that defect.

Three 200,000-permutation gate constructions are now reported; all permutation p-values use the add-one convention, $\hat{p} = (r+1)/(B+1)$, with exact Monte Carlo intervals. Under the historical seat-level gate, p13 remains the maximum at +0.4167, with familywise $p = 0.059230$, Monte Carlo 95% interval [0.058194, 0.060268]. Under the percentile episode-cluster-bootstrap sensitivity, p13 also remains the maximum and $p = 0.043455$, interval [0.042561, 0.044353]. The exact projection is the conservative reference because it has finite-sample coverage for the discrete episode mean; the percentile bootstrap is retained symmetrically but has no comparable small-sample coverage guarantee. Under the conservative exact-episode gate, p13 is ineligible: its low-δ lower bound falls below 0.05 and its high-δ upper bound exceeds 0.95. Only p04/s2p and p05/s2a pass both gates; the largest eligible slope belongs to p05/s2a (+0.0833), with familywise $p = 0.773206$, interval [0.771363, 0.775039].

For p13/s2a, the percentile bootstrap admitted both conditions as interior—δ=.10: [0.083, 0.667]; δ=.90: [0.583, 0.917]—whereas the conservative exact projection rejected both—[0.047, 0.800] and [0.287, 0.954], respectively. Neither recorded cell was at an exact corner. The eligibility difference therefore arises from small-sample interval width and coverage behavior, not from a corner interval falsely passing the gate.

The complete data-dependent gate is dynamically reapplied within every permutation, not frozen from the observed-data mask. The implementation precomputes 56 possible-composition gate values and performs 25,600,000 condition-gate lookup applications at B=200,000. In a deliberately incorrect comparison that froze the observed-data mask, the maximum statistic differed from the dynamic procedure in 718 of 5,000 null draws (14.4%), showing that reapplication materially changes the reference distribution. Lookup/direct parity and the regression are recorded in `docs/analysis/submission/round5/round5-review-audit.md`.

An exhaustive attainability audit shows that, with six episodes per condition, 12 of the 28 possible episode-value compositions pass the exact gate. Their sample means range from 0.333 to 0.667, but eligibility depends on the full $0, 0.5, 1$ composition rather than on the mean alone. Two eligible cells can therefore differ by at most 0.333. Under the archived 32-candidate null structure, the add-one familywise tail probability at that maximum attainable slope is 0.075040, with Monte Carlo 95% interval [0.073884, 0.076198]; no exact-gate result in this archived family can reach 0.05. The exact analysis is therefore a valid dependence-aware sensitivity but not a powered disconfirmation of a p13-sized capability claim. The record neither prospectively confirms nor decisively disconfirms p13; it identifies a replication target whose next test must be sized prospectively (Appendix A.4). A Phase 6 test will preregister the candidate family, episode-level dependence unit, interiority gate, maximum statistic, familywise decision rule, and sample size before any data are collected. A Phase 6 test will preregister the candidate family, episode-level dependence unit, interiority gate, maximum statistic, familywise decision rule, and sample size before any data are collected.

These variants were specified and executed together only after external review exposed the original defects. No familywise gate was registered at the original freeze, and the repository contains no seal-before-compute record for these sensitivities. The analyses ran in GitHub Actions against archived databases with fixed seeds, and their outputs were committed regardless of direction. Accordingly, the favorable $p = 0.043455$ variant cannot create prospective confirmation, just as the other variants cannot retroactively strengthen the frozen claim. The historical mechanical P5-3 verdict remains visible because clause (b) fired strongly and p13 passed the rule as frozen. Scientifically, p13 is a replication target, not a finding. Clause (a) did not prospectively establish the capability claim because the frozen search lacked family control; the conservative post-adjudication procedure is too underpowered at n=6 to provide decisive evidence against it.

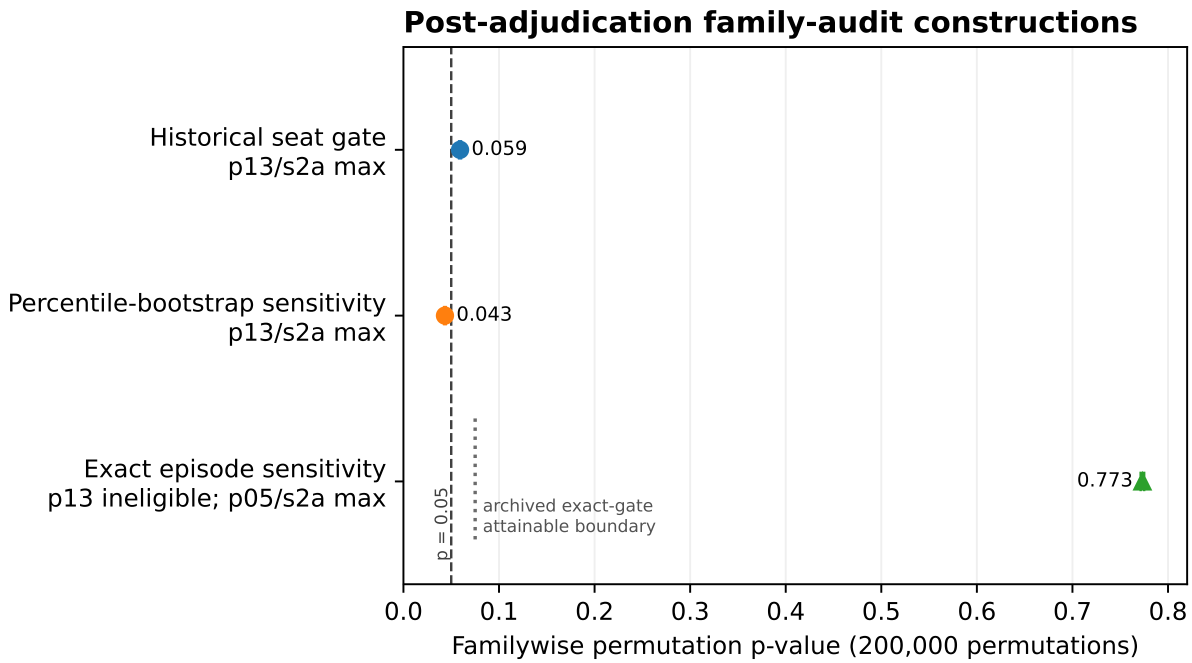

Figure 5. Post-adjudication familywise constructions. The first two points are p13/s2a under the historical and percentile-bootstrap gates. Under the conservative exact-episode gate, p13 is ineligible; the third point is the familywise result for the largest eligible candidate, p05/s2a. The dotted line at $p = 0.075040$ marks the estimated minimum attainable familywise p-value for the archived $n = 6$, 32-candidate exact-gate design and applies only to that construction. None of the procedures was registered at the original freeze; no point creates prospective confirmation.

4.5 Auditability

Twelve registered author predictions were refuted by data and published. Four underspecified analysis choices were resolved outcome-blind. A two-sided gate blocked a false ceiling conclusion. Sentinels detected a time-indexed behavioral discontinuity in an unversioned endpoint and exposed a monitoring gap that was repaired with an attestation gate. The public capsule now verifies all 4,916 confirmatory Phase 3–5 runs with zero live model calls: 4,896 LLM runs replay byte-exact and 20 deterministic baselines are independently recomputed. External review then found the family-error and dependence defects discussed above; the archived record made both diagnosable and correctable.

The claim is bounded: the machinery is procedurally exact and extensively auditable. Its strongest achievement is not self-validation but preservation of enough provenance for outsiders to identify where procedural correctness stopped short of statistical validity.

5. Discussion

5.1 Implications

For synthetic-participant practice, broad marginal resemblance is a weak validation target. Lightweight conditioning can generate plausible aggregate levels and substantial cross-prompt dispersion while yielding small observed treatment-response point estimates whose uncertainty remains wide. Validation should therefore report response surfaces over declared interventions and representation families, together with assay-sensitivity checks, dependence-aware uncertainty, model/provider provenance, and temporal monitoring. Statistical calibration [Hullman et al. 2026], causal-surrogacy assumptions [Persson et al. 2026], and latent-drift diagnostics [Lin et al. 2026] are complementary rather than competing safeguards.

The result can be read as a synthetic-subject analogue of the reduced-form/structural distinction associated with the Lucas critique [Lucas 1976]: fitting or selecting for aggregate resemblance need not identify behavior under a changed treatment. In psychometric terms, it is a construct-validity and assay-sensitivity problem [Cronbach & Meehl 1955; ICH E10; Temple & Ellenberg 2000]. Agent-based modeling's equifinality and pattern-oriented validation provide a related analogy [Windrum et al. 2007; Grimm et al. 2005]. These are organizing analogies, not claims that the study literally estimates a structural economic model or validates a human measurement model.

The results also show why binary certification language should be used sparingly. The historical P5-1a predicate passes under its frozen seat-level rule and under the conservative episode-exact interval, but fails under a reasonable Dirichlet–Jeffreys sensitivity. The underlying continuous evidence is clearer than the thresholded label: plug-in estimates condition on empirical boundary concentration and therefore tend upward, whereas the Jeffreys posterior shrinks six-episode corner cells toward the interior and therefore tends downward. These views bracket the composition claim under opposite conditioning choices rather than compete as interchangeable estimates; the posterior medians remain above one-half while the lowest 95% bound crosses it.

For deployment, behavior that can be rewritten by a sentence, an action token, or an identity prefix is safety-relevant. The label-conflict result does not show that lexical valence always dominates incentives: numerical payoffs move behavior in other cells, and the observed choice may reflect semantic framing, learned game-theoretic priors, or their interaction.

5.2 The precommitted discussion, and what changed after review

The Phase 5 discussion was sealed before its data existed. Excerpt (full text in supplement; sha `1f1d7de9…e356`):

> "The headline of Phase 5 is an existence result the program registered against itself: at least one persona in the sealed sixteen passed the two-sided assay gate and showed the registered signature of incentive sensitivity. The author's registered prediction—that none would—is refuted, and the refutation is the finding. … The capability was recoverable by content-side conditioning… The scope of the claim is deliberately narrow."

Precommitted interpretation	Current status after external review and zero-call reanalysis
At least one persona establishes an unconfounded incentive-response existence result	**Not prospectively established.** The frozen search lacked family control, and the post-adjudication procedures were method-dependent and unregistered. The conservative exact procedure cannot reach conventional familywise rejection in the archived n=6 design. See §4.4; p13 remains a replication target.
"No game-relevant instruction—trait words only"	Restated precisely: no explicit game terminology, action recommendation, or payoff reference; traits are strategically relevant information, and name, age, and occupation are uncontrolled semantic treatments.

Precommitted interpretation	Current status after external review and zero-call reanalysis
Persona framing “contested or beat” task switches	Pooled choice result survives exact episode inference, but every repeated conflict subcell is mixed; the pooled dominance classification is entirely carried by the word/payoff-confounded swap cell.
Bare corners characterize the configuration rather than the model’s capability envelope	Retained only as a future hypothesis. The p13 evidence no longer supports it; clause (b) demonstrates a robust choice reversal but does not identify incentive sensitivity.

The sealed text remains unchanged because its evidentiary value lies partly in making interpretive error visible. The correction is additive and explicit rather than silently rewritten.

6. Limitations

The confirmatory predicates were registered and adjudicated separately at nominal thresholds; no study-wide alpha allocation or familywise rule was registered across P5-1a, P5-1b, P5-2, P5-3(a), P5-3(b), and the sequential X1 extension. They address distinct estimands and are not interpreted here as one omnibus test, but study-level false-positive exposure is consequently greater than under a prospectively hierarchical or alpha-spending design. A Phase 6 replication should preregister the primary/secondary hierarchy, candidate families, dependence units, maximum statistics, and cross-predicate error allocation before data collection.

The primary evidence comes from one deployment and sixteen complete prompt bundles. Generalization to a persona generator, other models, or human participants is not identified. The uncertainty views answer different questions: the latent-propensity posterior is prior-dependent, the plug-in/conditional bootstrap conditions on recorded boundary concentration, and the two-stage bootstrap changes the estimand by resampling prompts.

The treatment-response intervals are wide because each prompt-cell has six independent episodes and the exact projection retains uncertainty at empirical corners. The binary interior census is correspondingly sensitive to small-n discrete interval width. Explicit persona strings are paired across conditions, but latent-person invariance is untested. The continuation process was manipulated together with its textual representation. The persona-prefix contrast lacks a format-matched neutral control. The label-swap result cannot distinguish semantic valence from memorized game-theoretic associations because no non-PD control retained the same labels.

The registered choice-entropy secondary is defined and reported in Appendix A.1. Its matched-lattice decline survives composition matching, but unequal seat counts imply slightly different finite-sample plug-in bias across temperatures; the comparison is exploratory, mechanism-free, and does not isolate temperature from persona conditioning. The high-temperature continuation interaction was not registered, and the Gemini tier is descriptive under endpoint non-stationarity.

Human references are published and protocol-nonmatched. The original familywise analyses were specified after review and cannot create retrospective confirmation. The exact n=6 family is underpowered by construction; p13 remains a replication target, not evidence for or against a general capability envelope.

7. Reproducibility and data availability

The public repository contains the event stores, prompt registries, sealed registrations, adjudication records, timestamp proofs, post-adjudication analyses, figures, manuscript history, and review record. The one-command capsule verifies all 4,916 confirmatory Phase 3–5 runs with zero credentials and zero live model calls. The audit covers 320 registered Phase 3/X1 LLM runs, 20 deterministic Phase 3 baselines, 2,864 Phase 4 runs, and 1,712 Phase 5 runs; three additional completed legacy entry/diagnostic runs are also replayed but are not counted as confirmatory. Phase 3 replay re-renders every prompt, requires recorded-cache hash hits, reparses raw completions, recomputes actions, payoffs, and RNG draw counts, and checks recorded call parity. The deterministic P3-C3 baseline is independently recomputed from archived seeds and game objects.

Phase 3 used a legacy provider path without Phase 4–5 response IDs or deterministic request-body SHA capture. Phase 4–5 contain 30,421 normal request events and 30,397 response events; the 24-event difference is the disclosed provider-failure partial set. Each partial belongs to a failed run, was excluded from completed-run analyses and the 4,916-run replay denominator, and was never decoded as an action. Request attempts remain represented in request and budget accounting rather than being silently discarded; only completed replacement runs, where present under the registered replacement procedure, enter analysis. Those response records contain rendered prompts, bundle and request-body hashes, engine commit and provider route, raw text, and provider response IDs. Individual completion payloads were not provider-attested or separately hash-chained at receipt. Capsule checksum manifests and external timestamps make the released database snapshot tamper-evident relative to publication; replay cannot prove that no alteration occurred before snapshot sealing.

Reproduce the confirmatory record with:

```bash
git clone https://github.com/yoheinakajima/synthetic-players
cd synthetic-players/capsule
bash verify.sh
```

All post-adjudication analyses are zero-call scripts over the archived databases and are labeled separately from prospectively registered results.

8. Attribution

The human author selected the research questions, approved the registered designs, adjudicated which reviewer recommendations to adopt, and accepts responsibility for every claim. The autonomous pipeline executed registration, dispatch, adjudication, and replay as apparatus. AI reviewers supplied adversarial analysis that materially changed the manuscript, including the family-error diagnosis, the condition-mean identity, the dependence-unit critique, and the latent-person-invariance correction.

The role of one round-2 reviewer subsequently expanded from critique to specification of the post-adjudication analyses and management of their integration. Those analyses were executed by GitHub Actions against the archived databases; resulting commits were authored by Yohei Nakajima or the Actions bot. A later reviewer independently checked the branch through an anonymous blobless clone against the generated JSON and commit history. Review prompts, role changes, adopted recommendations, and the independent verification memo are archived under `docs/reviews/`. Venue-specific AI-assistance language will be conformed at submission.

Appendix A — Supplementary scope and prospective design

A.1 Temperature secondary

Choice entropy is defined as base-2 Shannon entropy, $H = -\sum_a p(a)\log_2 p(a)$, over round-one payoff-role choices. The historical registered secondary pooled all valid choices at each temperature, but the T=0.7 and higher-temperature samples had different composition. The matched-sweep reanalysis uses only persona-cell lanes observed at all three temperatures:

temperature	matched units	seats	pooled Shannon entropy (bits)	mean within-unit entropy (bits)
0.7	13	544	0.8310	0.4484
1.0	13	284	0.7822	0.2566
1.3	13	284	0.7698	0.2877

The registered pooled decline is partly composition-confounded but survives on the identical sweep lattice. Mean within-unit entropy is reported separately because pooled entropy can remain high when different prompt-cell units occupy opposite boundaries. Neither statistic identifies a temperature mechanism. The plug-in Shannon estimates use unequal seat counts (544 versus 284 and 284), so their small finite-sample biases are not identical; no bias-corrected entropy claim is made.

A.2 Other supplementary findings

RPS retained a role-attached rock bias after neutral symbols and randomized order, with a cross-vendor sign reversal. The adversary suite showed opponent-contingent sequential structure. A sentinel case study documents endpoint drift and the resulting monitoring repair. These results and the Claude Haiku entry-gate failure remain in the public supplementary record but are outside the main causal arc.

A.3 Prospective replication

A Phase 6 replication will preselect one target or a small candidate family and preregister the complete familywise procedure: candidate set, episode-level unit, interiority gate, maximum statistic, decision threshold, and sample size. It should also include a format-matched neutral prefix, a continuation-probability × wording factorial, and a structurally equivalent non-PD label-conflict control. The registered power calculation must simulate the exact decision rule under its declared dependence model rather than reuse the archived 32-candidate search.

A.4 Research record

The complete correction ledger, sealed discussion text, dead-predictions ledger, reviewer-role disclosures, and mechanical v11⟶v12 disposition matrix are maintained in `docs/reviews/` and `docs/analysis/submission/`. Historical artifacts are never silently rewritten; current interpretations are linked to the versions they amend.

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